

Wednesday, Jan 21

$$1) F = (A+C+D)(A+C+\bar{D})(C+D)$$

$$\text{let } (A+C) = x$$

$$(A+C+0)(A+C+\bar{D})(C+0)$$

$$(x+D)(x+\bar{D})(C+D)$$

$$(x+D)(x+\bar{D}) = x \quad \text{Identity}$$

$$(x)(C+D)$$

$$(A+C)(C+D) \quad \because x = A+C$$

$$AC + AD + CC + CD$$

$$C + \underline{AC + CD + AD} \quad (\text{Absorption law})$$

$$F = C + AD$$

$$Q2) F = x\bar{y} + xyz + yz$$

$$x\bar{y} + (xyz + yz)$$

$$x\bar{y} + yz(x+1) \quad \text{Factorised}$$

$$x\bar{y} + yz \quad (x+1) = 1 \quad \text{Null Law}$$

$$F = x\bar{y} + yz$$

$$Q3) (A+C+D)(A+C+\bar{D})(A+\bar{C}+D)(B+C)$$

$$\text{let } A+C = x$$

$$(x+D)(x+\bar{D})(A+\bar{C}+D)(B+C)$$

$$(A+C)(A+\bar{C}+D)(B+C) \quad (x+D)(x+\bar{D}) = x$$

$$(A+C(\bar{C}+D))(B+C) \quad \text{Factorise } A$$

$$(A+C\bar{C}+CD)(B+C) \quad C\bar{C} = 0$$

$$(A+CD)(B+C)$$

$$AB + AC + BCD + CD \quad \text{Absorption law } BCD + CD \rightarrow CD$$

$$F = AB + AC + CD$$

